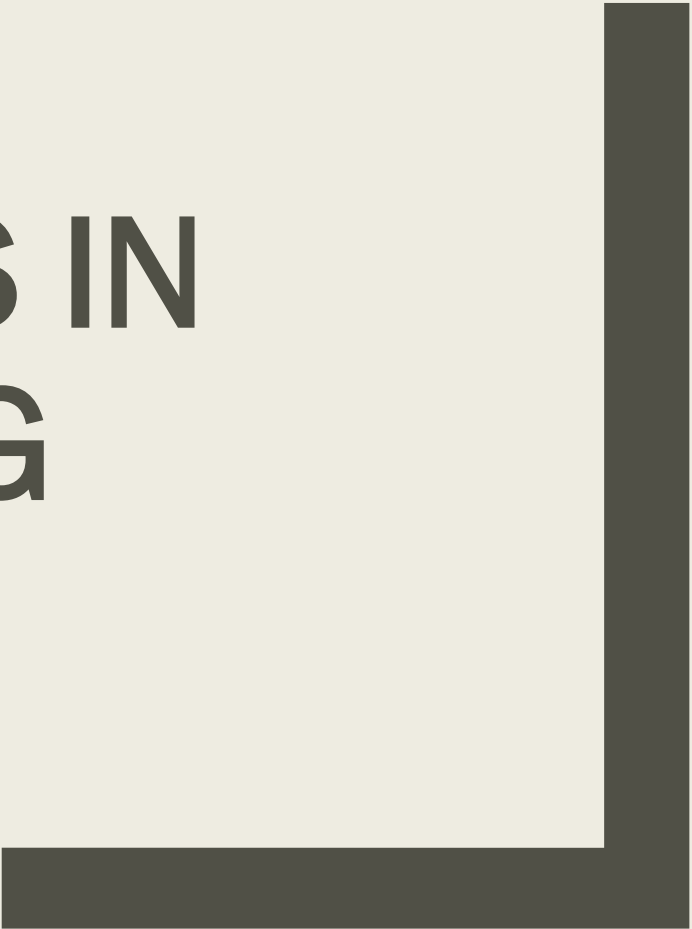




NEW HORIZONS IN ENGINEERING

Bobir Muratov

Introduction

- Need for change in the education of engineers stemming from the increasing tempo of technological change and scientific understanding

Introduction

- What are the grand challenges for engineering in our country today?

NAEs Grand Challenges for Engineering

- Advance personalized learning
- Make solar energy economical
- Enhance virtual reality
- Reverse engineer the brain
- Engineer better medicines
- Advance health informatics
- Restore and improve urban infrastructure
- Secure cyberspace
- Provide access to clean water
- Provide energy from fusion
- Prevent nuclear terror
- Manage the nitrogen cycle
- Develop carbon sequestration methods
- Engineer the tools of scientific discovery

New Horizons in Engineering

- Environmental responsibility and sustainable development
- Autonomous vehicle development
- Internet of things, big data, and cyber security
- Other new horizons for engineering